Intelligent Airline Customer Satisfaction Analysis System Using Multi-Agent AI Workflow

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Dataset: https://www.kaggle.com/datasets/sjleshrac/airlines-customer-satisfaction

Student Name:

Student ID:

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1. Application and Workflow Selection

This part presents the choice of intelligent application and provides the reasons why it is applicable in the airline sector. It provides a reason to analyze customer satisfaction, the process of work related to a multi-agent workflow adoption, the system goals, and the possible effect of introducing the AI-powered decision-support system to real-life operations of an airline.

1.1 Introduction and Motivation of the Selected Application

The issue of customer satisfaction is a crucial factor in the airline industry's success since direct influence is evident on passenger loyalty, brand image, and revenues in the long-term perspective. With intense competition within the international aviation industry, airlines have to keep track and enhance the quality of services to keep up with the growing customer expectations. Contemporary airlines have gathered a lot of information concerning demographics of the passengers, travelling experience, ratings of various services, and operational efficiency. Nonetheless, manual analysis of such large-scale data is tedious and time-consuming and would not necessarily display complicated patterns that could affect the level of satisfaction (Julian and Botti, 2019).

The project is a proposal of an intelligent airline customer satisfaction analysis system that will be based on artificial intelligence and machine learning methods. Multi-agent workflow in the system automates the data preprocessing, classification, sentiment analysis, prediction, and visualization. The overall idea behind this application is the necessity to give airlines a scalable, precise, and information-based decision-support system that will provide the ability to proactively make improvements related to services as well as improve passenger experience and make strategic planning information-based (Amin et al., 2023).

1.2 Explanation of the Selected Workflow with Diagram

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```python
# Install Kaggle
!pip install -q kaggle

KAGGLE_USERNAME = "your_kaggle_username"
KAGGLE_KEY = "your_kaggle_api_key"

import os

# Set environment variables
os.environ["KAGGLE_USERNAME"] = KAGGLE_USERNAME
os.environ["KAGGLE_KEY"] = KAGGLE_KEY

# Create directory for dataset
os.makedirs("airlines_customer_satisfaction_dataset", exist_ok=True)

# Download the dataset
!kaggle datasets download -d sjleshrac/airlines-customer-satisfaction -p airlines_customer_satisfaction_dataset

# Unzip the dataset
!unzip -o airlines_customer_satisfaction_dataset/*.zip -d airlines_customer_satisfaction_dataset

# List downloaded files
os.listdir("airlines_customer_satisfaction_dataset")
```

```
Dataset URL: https://www.kaggle.com/datasets/sjleshrac/airlines-customer-satisfaction
License(s): CC0-1.0
Downloading airlines-customer-satisfaction.zip to airlines_customer_satisfaction_dataset
  0% 0.00/1.64M [00:00<?, ?B/s]
100% 1.64M/1.64M [00:00<00:00, 592MB/s]
Archive:  airlines_customer_satisfaction_dataset/airlines-customer-satisfaction.zip
  inflating: airlines_customer_satisfaction_dataset/Invistico_Airline.csv

['airlines-customer-satisfaction.zip', 'Invistico Airline.csv']
```

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```python
# Agent 1: Data Preprocessing & Cleaning with Corrected Plots

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder
```

```python
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay
import matplotlib.pyplot as plt
import seaborn as sns
```

```python
import pandas as pd
import random
from textblob import TextBlob
import matplotlib.pyplot as plt
import seaborn as sns
```

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

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```python
print("\n=== Dataset Info ===")
print(df.info())
print("\n=== Missing Values ===")
print(df.isnull().sum())

# Plot missing values heatmap
plt.figure(figsize=(10,6))
sns.heatmap(df.isnull(), cbar=False, cmap='viridis')
plt.title("Missing Values Heatmap")
plt.show()

# Encode categorical columns including target for numeric correlation
categorical_cols = ['Gender', 'Customer Type', 'Type of Travel', 'Class', 'satisfaction']
le = LabelEncoder()
for col in categorical_cols:
    df[col] = le.fit_transform(df[col])

# First 10 rows after encoding
print("\n=== First 10 rows after encoding categorical variables ===")
print(df.head(10))

# Plot distributions of numerical columns
numerical_cols = df.select_dtypes(include=['int64', 'float64']).columns
df[numerical_cols].hist(figsize=(15,10), bins=20, color='skyblue')
plt.suptitle("Distribution of Numerical Features", fontsize=16)
plt.show()

# Count plots for original categorical features (after encoding)
for col in ['Gender', 'Customer Type', 'Type of Travel', 'Class']:
    plt.figure(figsize=(6,4))
    sns.countplot(x=col, data=df)
    plt.title(f"Count Plot of {col}")
    plt.show()
```

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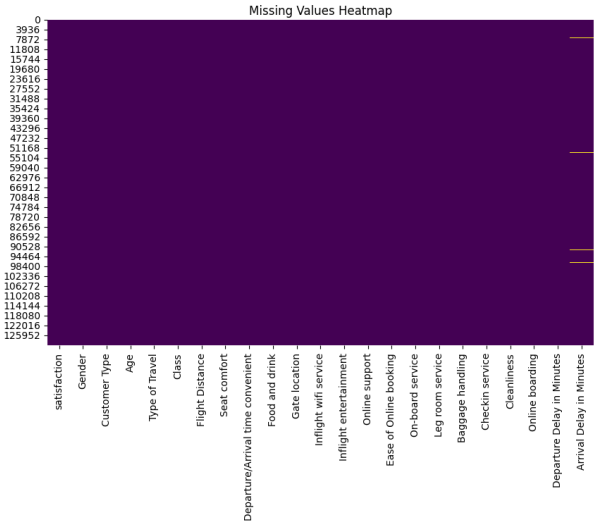

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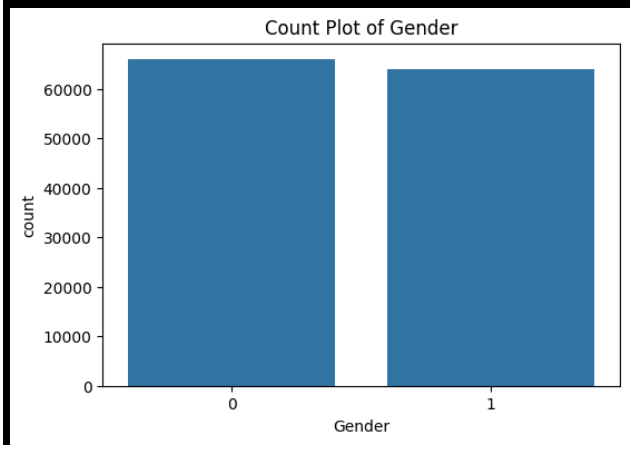

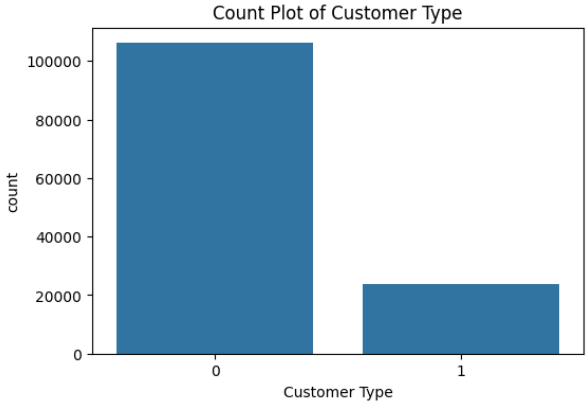

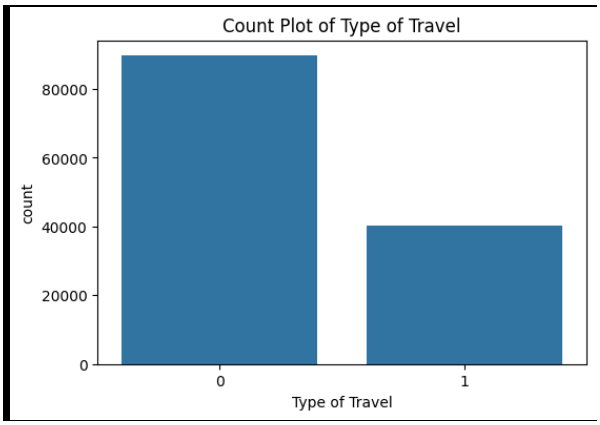

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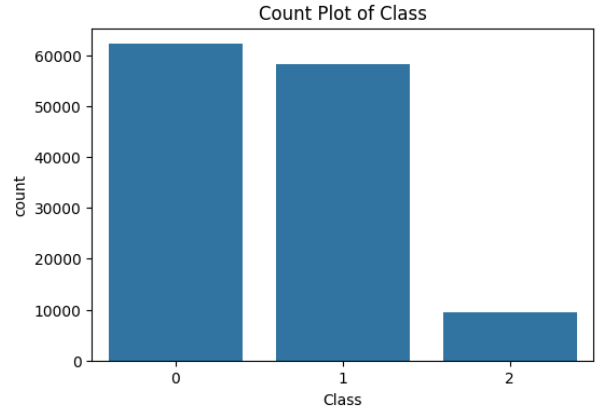

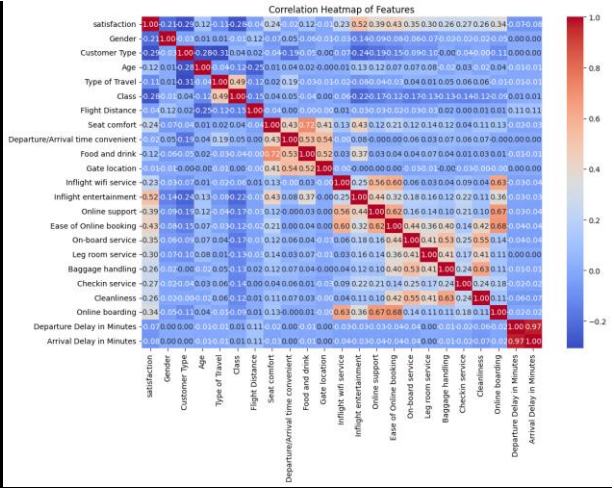

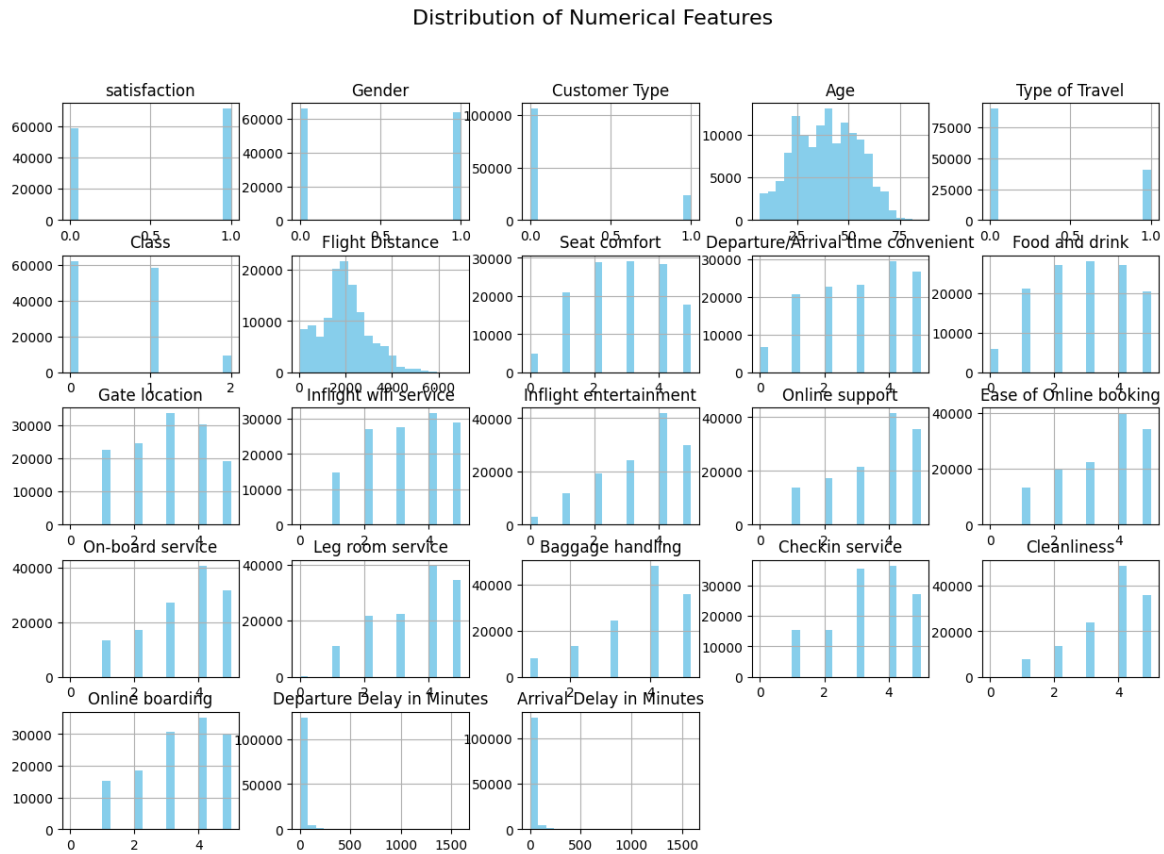


```
=== First 10 rows of the dataset ===
  satisfaction  Gender   Customer Type  Age   Type of Travel     Class  \
0    satisfied  Female  Loyal Customer   65  Personal Travel       Eco
1    satisfied    Male  Loyal Customer   47  Personal Travel  Business
2    satisfied  Female  Loyal Customer   15  Personal Travel       Eco
3    satisfied  Female  Loyal Customer   60  Personal Travel       Eco
4    satisfied  Female  Loyal Customer   70  Personal Travel       Eco
5    satisfied    Male  Loyal Customer   30  Personal Travel       Eco
6    satisfied  Female  Loyal Customer   66  Personal Travel       Eco
7    satisfied    Male  Loyal Customer   10  Personal Travel       Eco
8    satisfied  Female  Loyal Customer   56  Personal Travel  Business
9    satisfied    Male  Loyal Customer   22  Personal Travel       Eco

   Flight Distance  Seat comfort  Departure/Arrival time convenient  \
0              265             0                                  0
1             2464             0                                  0
2             2138             0                                  0
3              623             0                                  0
4              354             0                                  0
5             1894             0                                  0
6              227             0                                  0
7             1812             0                                  0
8               73             0                                  0
9             1556             0                                  0
...
Online boarding                      0
Departure Delay in Minutes           0
Arrival Delay in Minutes           393
dtype: int64
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
```

```
=== Summary Statistics ===
        satisfaction         Gender  Customer Type            Age  \
count  129880.000000  129880.000000  129880.000000  129880.000000
mean        0.547328       0.492616       0.183092      39.427957
std         0.497757       0.499947       0.386743      15.119360
min         0.000000       0.000000       0.000000       7.000000
25%         0.000000       0.000000       0.000000      27.000000
50%         1.000000       0.000000       0.000000      40.000000
75%         1.000000       1.000000       0.000000      51.000000
max         1.000000       1.000000       1.000000      85.000000

       Type of Travel           Class  Flight Distance   Seat comfort  \
count   129880.000000  129880.000000    129880.000000  129880.000000
mean         0.309416       0.593864      1981.409055       2.838597
std          0.462255       0.621378      1027.115606       1.392983
min          0.000000       0.000000        50.000000       0.000000
25%          0.000000       0.000000      1359.000000       2.000000
50%          0.000000       1.000000      1925.000000       3.000000
75%          1.000000       1.000000      2544.000000       4.000000
max          1.000000       2.000000      6951.000000       5.000000

       Departure/Arrival time convenient  Food and drink  ...  Online support  \
count                       129880.000000   129880.000000  ...   129880.000000
mean                            2.990645        2.851994  ...        3.519703
...

[8 rows x 23 columns]

Cleaned dataset saved as 'airline_cleaned.csv'.
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
```

```
=== First 10 rows after encoding categorical variables ===
   satisfaction  Gender  Customer Type  Age  Type of Travel  Class  \
0             1       0              0   65               1      1
1             1       1              0   47               1      0
2             1       0              0   15               1      1
3             1       0              0   60               1      1
4             1       0              0   70               1      1
5             1       1              0   30               1      1
6             1       0              0   66               1      1
7             1       1              0   10               1      1
8             1       0              0   56               1      0
9             1       1              0   22               1      1

   Flight Distance  Seat comfort  Departure/Arrival time convenient  \
0              265             0                                  0
1             2464             0                                  0
2             2138             0                                  0
3              623             0                                  0
4              354             0                                  0
5             1894             0                                  0
6              227             0                                  0
7             1812             0                                  0
8               73             0                                  0
9             1556             0                                  0
...
8                      0.0
9                     26.0

[10 rows x 23 columns]
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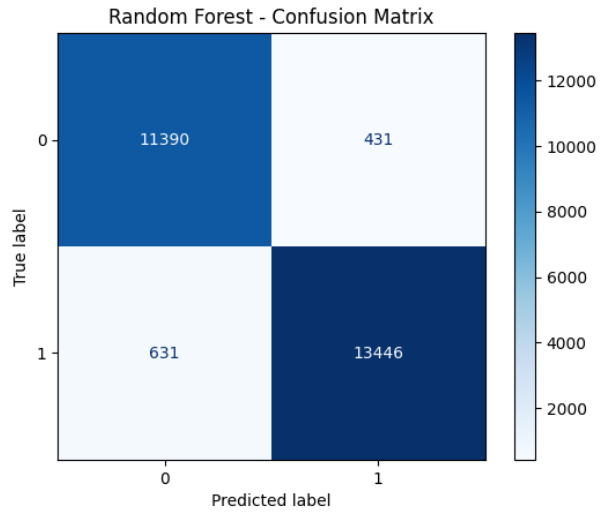

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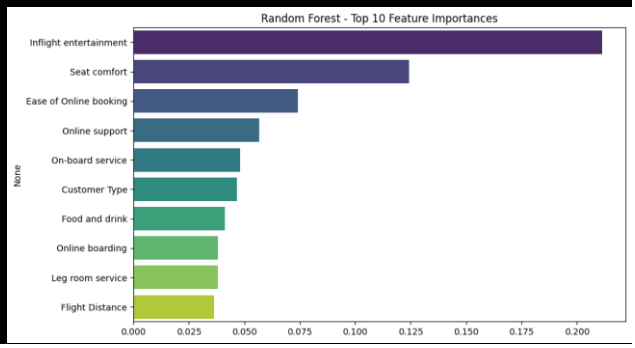

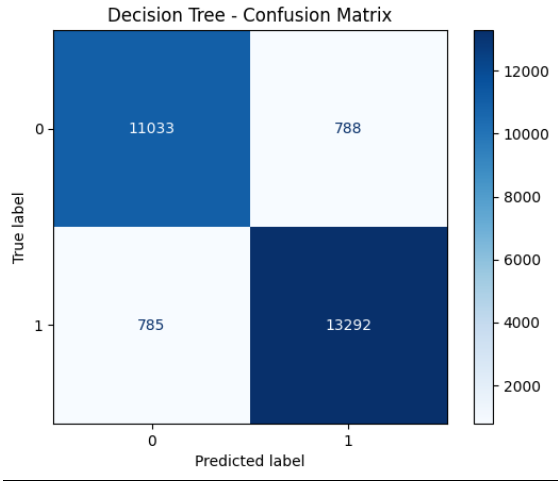

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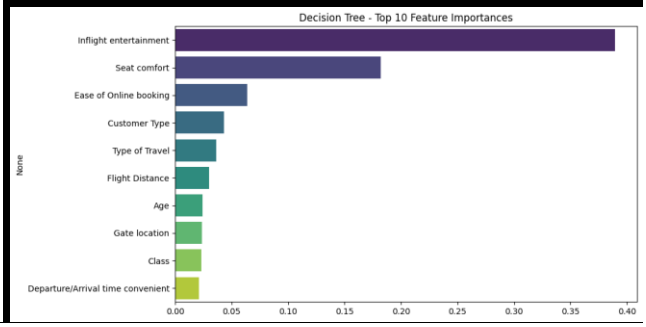

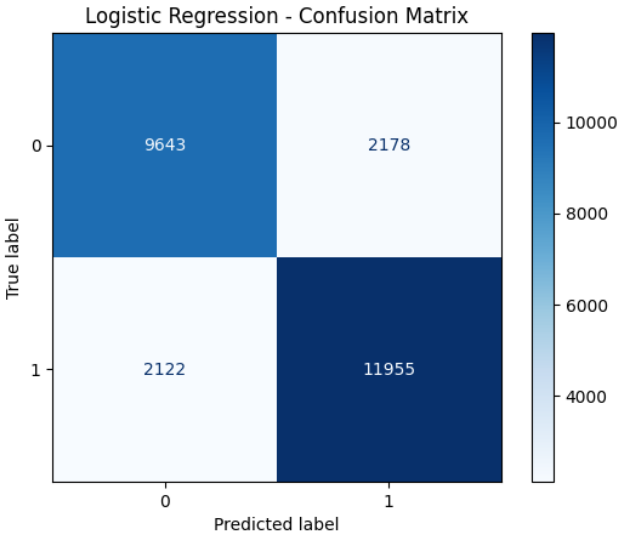

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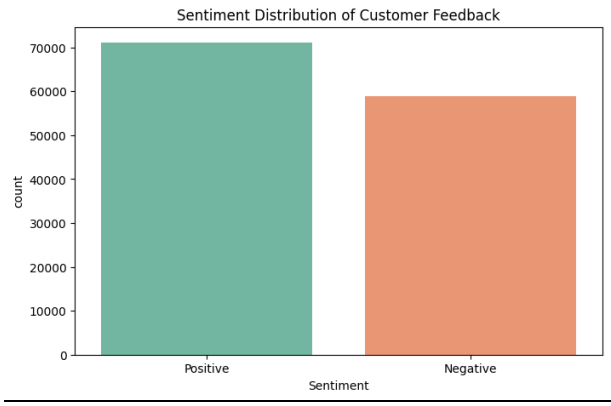


```
  satisfaction                                Feedback Sentiment
0    satisfied  Very satisfied with the airline services.  Positive
1    satisfied       Everything was smooth and enjoyable.  Positive
2    satisfied      Great service and comfortable flight!  Positive
3    satisfied            Loved the in-flight experience.  Positive
4    satisfied      Great service and comfortable flight!  Positive
5    satisfied  Very satisfied with the airline services.  Positive
6    satisfied  Very satisfied with the airline services.  Positive
7    satisfied            Loved the in-flight experience.  Positive
8    satisfied       Excellent staff and on-time service.  Positive
9    satisfied       Everything was smooth and enjoyable.  Positive
/tmp/ipython-input-2704426512.py:48: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in

  sns.countplot(x='Sentiment', data=df, palette='Set2')
```

```
=== Actionable Recommendations ===
- Improve 'Inflight entertainment' to significantly increase customer satisfaction.
- Improve 'Seat comfort' to significantly increase customer satisfaction.
- Improve 'Ease of Online booking' to significantly increase customer satisfaction.
- Improve 'Online support' to significantly increase customer satisfaction.
- Improve 'Customer Type' to significantly increase customer satisfaction.

General Recommendations:
- Reduce arrival and departure delays.
- Improve seat comfort and leg room services.
- Enhance online boarding and check-in experience.
- Maintain cleanliness and in-flight service quality.
```

```
Missing values in target: 0

=== Agent 4: Prediction Results ===
Accuracy: 0.9601

Classification Report:

              precision    recall  f1-score   support

           0       0.95      0.96      0.96     11821
           1       0.97      0.96      0.96     14077

    accuracy                           0.96     25898
   macro avg       0.96      0.96      0.96     25898
weighted avg       0.96      0.96      0.96     25898
```

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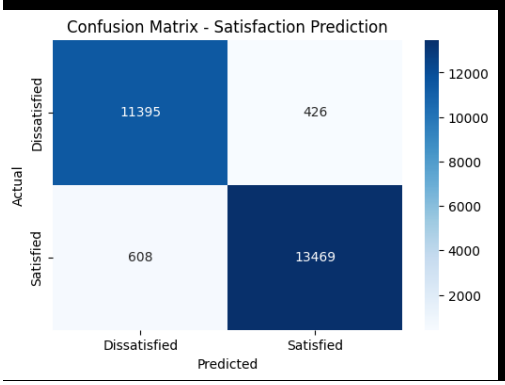

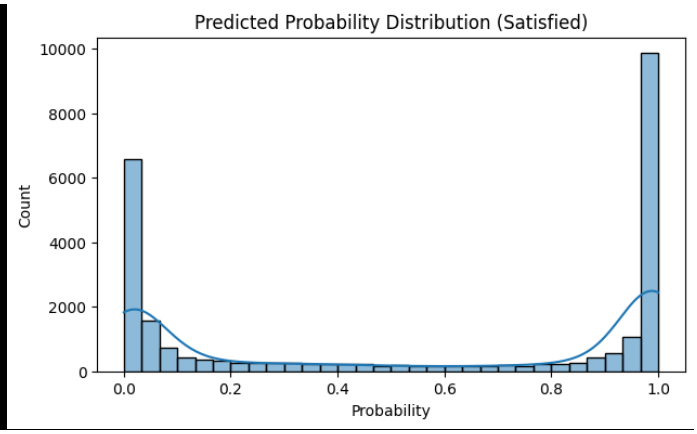

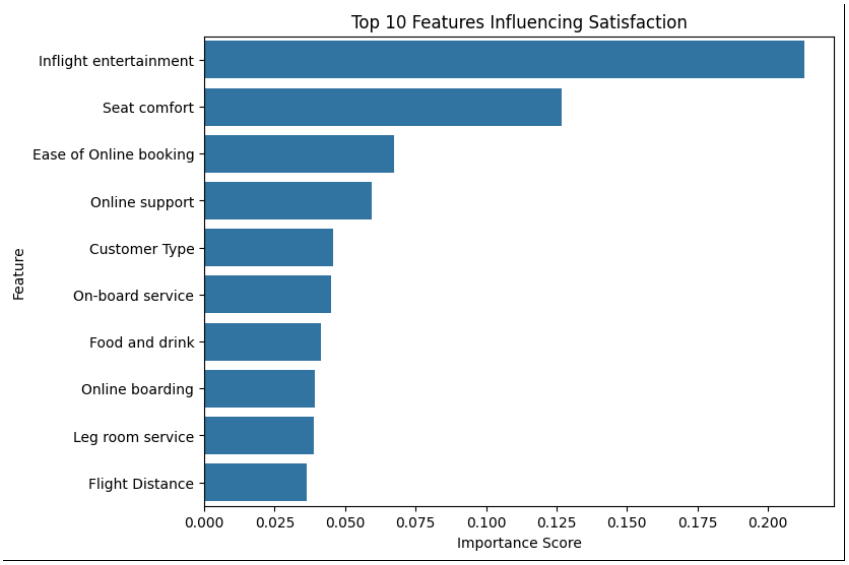

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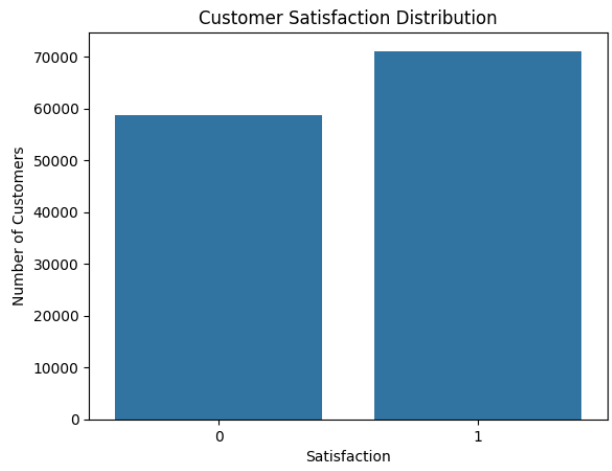

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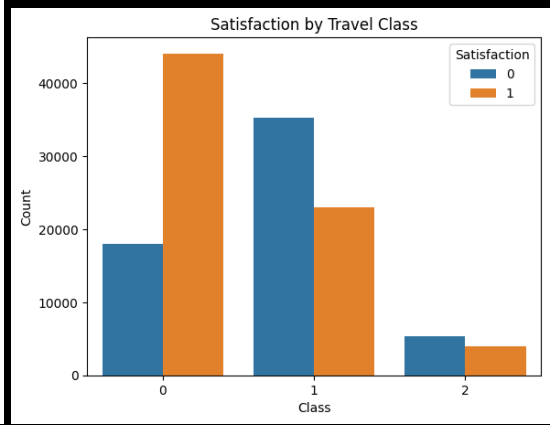

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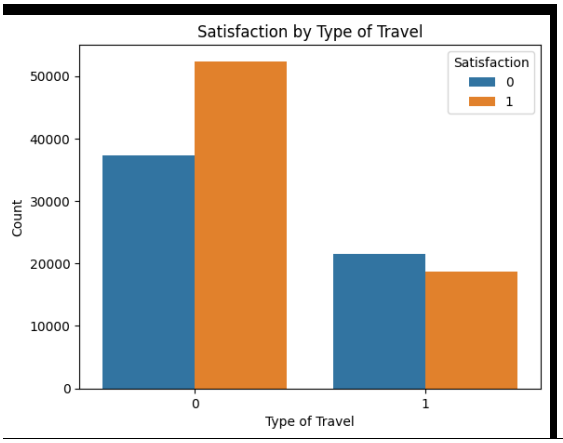

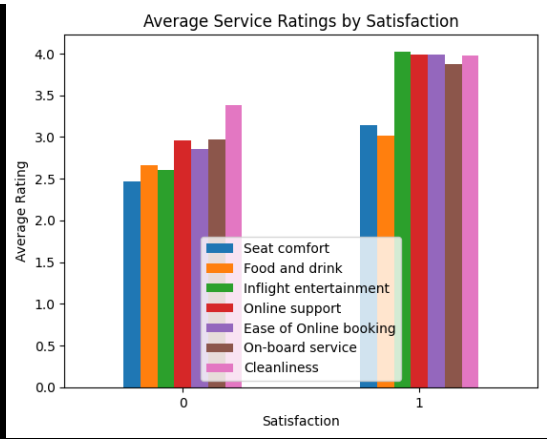

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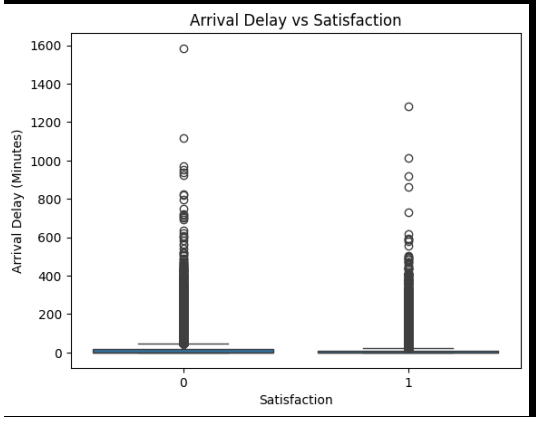

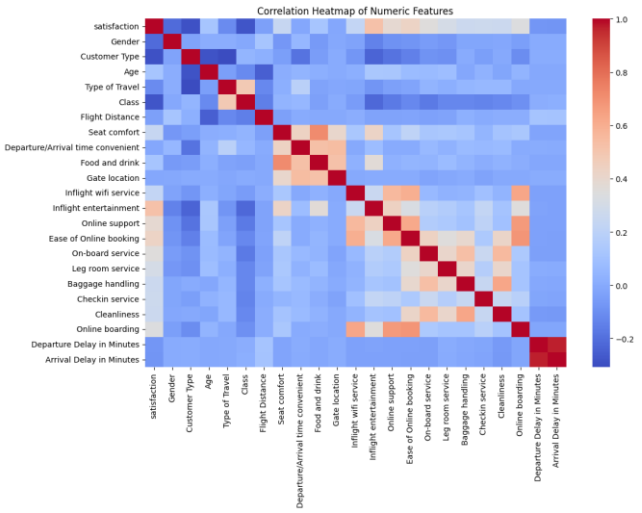

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